Berge–Fulkerson Covers, Cycle Double Covers, Flows and Exact Normality Defects of the First Counterexamples to the Petersen Coloring Conjecture

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CAOS Research program, github.com/fsantibanezleal/CAOS_RESEARCH

2026-09-03

Version 0.01

Preprint (independent certification and consequence audit) · v0.01, 3 September 2026 ·

CAOS Research open-problems programme · problem `petersen-coloring`, experiments 001–004

Version DOI 10.5281/zenodo.DOIVERSION · concept DOI (always latest) 10.5281/zenodo.DOICONCEPT

Not peer reviewed. Source, declared hypotheses, artifacts and verdicts: github.com/fsantibanezleal/CAOS_RESEARCH

Provenance. The counterexamples are due to Putman (two graphs on 112 vertices, August 2026), with a human-checkable proof by Jooken, and to Goedgebeur, Jooken, Máčajová, Mattiolo and Mazzuoccolo (52 vertices and infinite families). This paper contributes an independent certification and the audit of what the conjecture used to imply.

Abstract

Jaeger's Petersen coloring conjecture (1988), which asserted that every bridgeless cubic graph admits an edge map onto the Petersen graph sending every vertex star onto a vertex star, was refuted in August 2026: Putman exhibited two nonisomorphic 112-vertex counterexamples with SAT-solver certificates, Jooken gave a human-checkable proof, and Goedgebeur, Jooken, Máčajová, Mattiolo and Mazzuoccolo found a 52-vertex counterexample together with infinite families, placing the smallest counterexample between 38 and 52 vertices. The conjecture had been the strongest known sufficient condition for the Berge–Fulkerson conjecture and the 5-cycle double cover conjecture. We audit, by exact proof-carrying computation, what survives on the three retrievable counterexamples. First, independent encodings sharing no variable scheme with the public ones refute Petersen colorings and normal 5-edge-colorings of all three graphs with proofs checked by `drat-trim`; five colorable controls are accepted, and Putman's four public proofs verify under our checker. Second, every one of the three graphs admits a Berge–Fulkerson cover, a Berge cover by five perfect matchings, a Fan–Raspaud triple, a 5-cycle double cover and a nowhere-zero 5-flow, all given as explicit witnesses re-verified from the graph alone; none admits a nowhere-zero 4-flow. Their perfect matching index is exactly 4, one below the Petersen graph's value. The two 112-vertex graphs have oddness 4 and resistance 3, whereas the 52-vertex graph has oddness 2 and resistance 2, every value certified from both sides. AUDITFOUR All computations are propositional with checked DRAT certificates or explicit witnesses; hypotheses were declared before each

run, and a refuted prediction of ours (oddness 2 for the 112-vertex graphs) is preserved. The paper decides nothing about the general conjectures and says so.

1 Introduction

1.1 The conjecture and its ladder

Let P denote the Petersen graph. For a cubic graph G, a *Petersen coloring* is a map $\sigma : E(G) \to E(P)$ such that for every vertex v of G there is a vertex w of P with $\sigma(\partial_G(v)) = \partial_P(w)$, where $\partial(\cdot)$ denotes the set of edges at a vertex; equivalently, the three edges at v map bijectively onto the three edges at w. A proper 5-edge-coloring is *normal* if every edge uv is *poor* ($|c(\partial u) \cup c(\partial v)| = 3$) or *rich* ($= 5$); the *normal chromatic index* $\chi'_N(G)$ is the least k admitting a normal k-edge-coloring.

Conjecture 1.1 (Jaeger 1988 [1])**.** *Every bridgeless cubic graph admits a Petersen coloring.*

Jaeger [2] proved that a cubic graph has a Petersen coloring iff it has a normal 5-edge-coloring (an explicit bijective form is Proposition 2.1 of [3]). The conjecture implies the Berge–Fulkerson conjecture (six perfect matchings covering every edge exactly twice) and the 5-cycle double cover conjecture [4, 6]; the Berge conjecture (perfect matching index at most 5) and the Fan–Raspaud conjecture (three perfect matchings with empty intersection) follow from Berge–Fulkerson. Every 3-edge-colorable cubic graph is trivially Petersen colorable, so the whole ladder concerns snarks. Exhaustive generation had verified the conjecture for every snark on at most 36 vertices [7] and for every weak snark on 36 vertices [8], so any counterexample has at least 38 vertices [5].

1.2 The disproof

On 6 August 2026 Putman [3] published a simple bridgeless cubic graph G_{112} on 112 vertices and 168 edges with no Petersen coloring and no normal 5-edge-coloring, certified by CaDiCaL proofs checked with `drat-trim`, together with a nonisomorphic D_3-symmetric counterexample H_{112}; both have girth 5 and edge- and vertex-connectivity 3. The construction composes copies of the 4-pole F (the Petersen graph minus the endpoints of one edge) with claw six-poles: $L = 4F + C$ and $G_{112} = 3L + C$. Jooken [4] gave a human-checkable proof through the Petersen-coloring sets of F and L and a 4-clique obstruction in the line graph of P. Goedgebeur, Jooken, Máčajová, Mattiolo and Mazzuoccolo [5] then gave a cyclically 4-edge-connected girth-5 counterexample G_{52} on 52 vertices, infinite cyclically 4-edge-connected families, and the frontier $38 \le n_{\min} \le 52$ for the smallest counterexample; they verified a strong normal 6-edge-coloring of one 112-vertex graph and left open whether every bridgeless cubic graph has a normal 6-edge-coloring (their Conjecture 6, attributed to Šámal) and whether cyclically 5-edge-connected counterexamples exist. A 68-vertex counterexample was announced on a social network without a retrievable edge list and is not treated here.

1.3 What this paper does

The three retrievable counterexamples are the first bridgeless cubic graphs for which the Petersen route to a Berge–Fulkerson cover or a 5-cycle double cover is closed. We therefore ask, for each of G_{112}, H_{112} and G_{52}, which of the conjectures below the top of the ladder still hold, and how far each graph is from being colorable. Every answer is exact: a positive answer is an explicit witness re-verified by a standard-library checker that reads only the graph, and a negative answer is a DRAT proof checked by `drat-trim`. Section 2 records the independent certification (a replication, labeled as such), Section 3 the perfect matching covers, Section 4 the cycle double covers, flows, oddness and resistance, Section 5 the normal 6-edge-colorings and the exact defects, and Section 6 the open questions. Hypotheses were declared and committed before each run; one of our committed predictions was refuted by the machine and is reported.

2 Independent certification

2.1 The graphs

We use Putman's normalized digest: the SHA-256 of the compact JSON serialization of the sorted edge list with $u < v$. The three graphs are G_{112} (digest `dc16cc18...e8b`), H_{112} (digest `0f2d8858...35c7`), both transcribed from Putman's CC0 archive, and G_{52}, transcribed from the appendix of [5] with vertices renumbered from 1..52 to 0..51 (digest `27db5d3b...`); the full edge lists are in the repository.

Proposition 2.1 (machine-verified)**.** *Each of G_{112}, H_{112}, G_{52} is simple, cubic, connected, has edge connectivity 3, girth 5, and cyclic edge connectivity exactly 4: no set of at most three edges is cycle-separating (exhaustive search over all edge subsets of size at most three), and an explicit cycle-separating 4-cut exists (edges $\{2, 6, 9, 12\}$ of the sorted edge lists of both 112-vertex graphs and $\{2, 3, 8, 14\}$ of G_{52}).*

2.2 Encodings

Our Petersen encoding uses only edge-image variables $y_{e,f}$ ($e \in E(G)$, $f \in E(P)$), one image per edge, with the constraint that adjacent edges of G receive distinct adjacent edges of P. This suffices because P is triangle-free: three pairwise adjacent distinct edges of a triangle-free graph form a star. Two units break symmetry, using the edge-transitivity of P and the transitivity of an edge stabilizer on the four edges adjacent to a fixed edge. Our normal-5 encoding uses edge-color variables $x_{e,c}$, side-presence variables $p_{e,c}, q_{e,c}$ ("color c appears among the two other edges at u", respectively at v), and a rich indicator r_e; richness is $\neg(p_{e,c} \wedge q_{e,c})$ for every c and poorness is $p_{e,c} \leftrightarrow q_{e,c}$ for every c. Neither encoding shares a variable scheme with Putman's vertex-witness and missing-pair encodings.

Theorem 2.2 (machine-verified, replication)**.** *G_{112}, H_{112} and G_{52} admit no Petersen coloring and no normal 5-edge-coloring. All six refutations, and the three Petersen refutations with symmetry breaking removed, carry DRAT proofs accepted by `drat-trim` (Table 1). The same encoders accept the Petersen graph, K_4, the prism and the flower snarks J_5, J_7 with checker-validated colorings. Putman's four archived proofs are accepted by our `drat-trim` binary against his archived formulas.*

3 Perfect matching covers

Theorem 3.1 (machine-verified)**.** *Each of G_{112}, H_{112}, G_{52} admits a Berge–Fulkerson cover, a Berge cover by five perfect matchings, three perfect matchings with empty intersection, and a cover by four perfect matchings. Consequently their perfect matching index equals 4.*

instance	variables	clauses	solve (s)	proof (bytes)	check (s)
G_{112} Petersen	2,520	73,250	65.6	98,426,474	43.7
G_{112} normal-5	2,688	11,088	1,216.8	786,818,036	900.5
H_{112} Petersen	2,520	73,250	29.5	95,632,373	24.5
H_{112} normal-5	2,688	11,088	977.2	868,156,862	1,293.5
G_{52} Petersen	1,170	34,010	1.8	4,094,201	1.2
G_{52} normal-5	1,248	5,148	178.8	226,490,830	231.8

Table 1: Certified refutations (CaDiCaL 1.7.3 under WSL, `drat-trim`); every check printed `s VERIFIED`.

Proof. The matchings are explicit (the repository file `witnesses.json` of experiment 002 lists them as edge index sets) and were re-verified by a checker that tests each set to be a perfect matching and then counts the coverage of every edge. The index is at least 4 because the graphs are not 3-edge-colorable (Theorem 2.2 and the trivial direction of the conjecture). □

The value 4 is one below the Petersen graph's own perfect matching index 5, which our control recovers with a verified refutation of the four-matching instance. Thus, on the first counterexamples, the Berge–Fulkerson, Berge and Fan–Raspaud conjectures all hold, and the graphs are in this sense better behaved than the Petersen graph.

4 Cycle double covers, flows, oddness and resistance

Theorem 4.1 (machine-verified). *Each of G_{112}, H_{112}, G_{52} admits a 5-cycle double cover (five even subgraphs covering every edge exactly twice) and a nowhere-zero $\mathbb{Z}_5$-flow, and none admits a nowhere-zero $\mathbb{Z}_4$-flow. The oddness and resistance are*

$$\mathrm{odd}(G_{112}) = \mathrm{odd}(H_{112}) = 4, \quad \mathrm{odd}(G_{52}) = 2, \qquad \mathrm{res}(G_{112}) = \mathrm{res}(H_{112}) = 3, \quad \mathrm{res}(G_{52}) = 2.$$

Proof. Double covers and flows are explicit witnesses; the 4-flow refutations carry checked proofs. Oddness is decided from both sides: for a bound b, the formula asks for a perfect matching M and a vertex 2-coloring of the 2-factor $E \setminus M$ with at most b monochromatic edges. An odd cycle forces at least one monochromatic edge and an even cycle admits none, so the least satisfiable b is the oddness. Bounds $1, 2, 3$ are refuted with checked proofs for the 112-vertex graphs and bound 4 is satisfied by a perfect matching whose 2-factor has exactly four odd cycles (counted by the checker); for G_{52} bound 1 is refuted and bound 2 satisfied. Resistance uses a deletion indicator per edge and a proper 3-edge-coloring of the remaining edges, with the same two-sided protocol. □

Our committed prediction was oddness 2 for all three graphs; it was refuted for the 112-vertex graphs. The value is consistent with the theorem that the smallest snarks with oddness at least 4 and cyclic connectivity 4 have order 44 [8]. The 52-vertex graph, by contrast, has the least possible oddness of a snark.

5 Normal 6-edge-colorings and exact defects

AUDITFOURSECTION

6 Open questions and non-claims

Nothing here bears on the general Berge–Fulkerson, cycle double cover, 5-flow or normal 6-edge-coloring conjectures: the audit concerns three graphs. The questions left open by the disproof, the smallest counterexample in [38, 52], cyclically 5-edge-connected counterexamples, and Conjecture 6 of [5], remain open. The 68-vertex graph announced on a social network is not covered. Every claim above is traceable to an experiment folder with a declared hypothesis, a deterministic runner, artifacts with SHA-256 digests, and a verdict, under `problems/combinatorics/petersen-coloring/` in the repository.

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